

Vanitha Penchala

DATA ANALYST

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SUMMARY

- Experienced Data Analyst with 5+ years of expertise in utilizing advanced analytics and machine learning techniques to drive actionable insights and optimize business processes in healthcare and financial sectors.
- Skilled in data cleaning and preprocessing using Python (pandas, numpy) and SQL, significantly improving data quality and ensuring robust data management and integrity.
- Expert in developing and deploying machine learning models for predictive analytics and customer behavior analysis, including Random Forest, XGBoost, and collaborative filtering, achieving up to 88% accuracy in predictive models.
- Experienced in creating and managing interactive dashboards with Tableau and Power BI, providing real-time visualizations and actionable insights that increased stakeholder engagement and data-driven decision-making.
- Demonstrated success in leveraging cloud platforms such as AWS and Google Cloud Platform for scalable data storage and processing, reducing infrastructure costs and handling large-scale data analytics efficiently.
- Proficient in implementing advanced data governance protocols using Collibra and ensuring compliance with HIPAA and other data privacy standards, achieving adherence to privacy and security regulations.
- Experienced in conducting A/B testing and statistical analysis to optimize business strategies and engagement initiatives, resulting increase in preventive care visits and increase in cross-sell opportunities.
- Strong background in statistical analysis and data visualization, with expertise in methodologies such as ANOVA, hypothesis testing, regression analysis, and time series forecasting, driving actionable insights and strategic planning.

SKILLS

Methodologies:	SDLC, Agile, Scrum, Waterfall
Programming Language:	Python, SQL, R
Packages:	NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, TensorFlow, Seaborn, dplyr, ggplot2
Visualization Tools:	Tableau, Power BI, Advanced Excel (Pivot Tables, VLOOKUP, VBA, Macros)
IDEs:	Visual Studio Code, PyCharm, Jupyter Notebook, IntelliJ
Database:	MySQL, PostgreSQL, SQL Server, Oracle
Cloud Platform:	Amazon Web Services (AWS), Google Cloud Platform
Other Technical Skills:	SISS, SSRS, Machine Learning Algorithms, ETL Tools, Statistics, ServiceNow, MapReduce, Snowflake, SAS, Hadoop, Spark, Alteryx, Probability distributions, Mathematics, Confidence Intervals, ANOVA, Advance Analytics, OLAP & OLTP, SAS, MS Office Suite (Microsoft Word, PowerPoint, MS Visio), Hypothesis Testing, Regression Analysis, Linear Algebra, Advance Analytics, Google big query, Data Mining, Data Visualization, Data warehousing, Data transformation, Data Storytelling, Big Data, Data Integration, Data Interpretation, Data Pipeline, Association rules, Clustering, Classification, Regression, A/B Testing, Forecasting & Modelling, Data Cleaning, Data Wrangling, Git, GitHub, JIRA
Soft Skills:	Effective Communication, Problem-Solving, Time Management, Stakeholder Management, Presentation Skills, Risk Management, Flexibility, Networking, Optimization Techniques, Verbal Communication, Analytical Thinking, Attention to Detail, verbal and written communication

EXPERIENCE

Data Analyst | Baylor Scott and White Health, TX

Feb 2022 – Present

- Led a patient readmission prediction project using advanced data analytics and machine learning, reducing 30-day readmissions by 18% across 52 hospitals, impacting over 200,000 patient discharges annually.
- Utilized Python (pandas, numpy, scikit-learn) and SQL for data cleaning and preprocessing. Developed custom algorithms for handling missing data and outliers, improving data quality by 25% and enhancing model performance.

- Designed and implemented ETL pipelines with Apache Airflow, integrating data from 15+ sources, including Epic EHR and claims databases. Processed and standardized over 500GB of daily patient data, achieving 99.9% data accuracy and completeness.
- Engineered a distributed machine learning pipeline using PySpark on a Hadoop cluster, deploying a Random Forest model with 85% accuracy in predicting high-risk patients.
- Created interactive Tableau dashboards for real-time monitoring of readmission rates, risk factors, and intervention efficacy, increasing stakeholder engagement by 40% and accessed by 500+ healthcare professionals daily.
- Optimized SQL queries and implemented indexing strategies, reducing report generation time from 30 minutes to 12 minutes, a 60% improvement in system efficiency.
- Designed and maintained a cloud-based data warehouse using Snowflake, integrating real-time and historical data from 10+ systems. Supported 500+ concurrent users and reduced data retrieval time.
- Developed A/B testing frameworks to evaluate patient engagement strategies, resulting in a 22% increase in preventive care visits for high-risk patients and potentially preventing 1,000+ readmissions annually.
- Leveraged AWS services (EC2, S3, Lambda) to create a scalable, cost-effective infrastructure for predictive analytics, reducing infrastructure costs by 35% and handling 3x the original data volume.
- Implemented data governance protocols with Collibra to ensure HIPAA compliance and data security, achieving 99.9% adherence to privacy standards for over 2 million patients.

Data Analyst | Deloitte, India

May 2018 – Jul 2021

- Developed and implemented a customer churn prediction and retention optimization system for PayEase, India's leading digital payments platform with over 50 million users.
- Conducted a comprehensive data audit, identifying 15+ relevant data sources including transaction logs, user activity, customer support interactions, and app usage metrics. Utilized Python and pandas to clean and preprocess 2.3TB of historical data, improving data quality by 42% as measured by a custom data quality score.
- Designed and implemented a robust ETL pipeline using Apache Airflow, integrating data from MySQL, MongoDB, and Kafka streams. Processed an average of 1.2 million daily transactions and 5 million user events, reducing data preparation time.
- Leveraged Google Cloud Platform (GCP) for scalable data storage and processing. Utilized BigQuery for large dataset storage and querying, and Dataflow for distributed data processing, analyzing over 100TB of historical data to uncover long-term user behavior patterns.
- Created a recommendation engine using collaborative filtering and matrix factorization, analyzing historical transaction data to suggest personalized offers. Increased cross-sell opportunities by 29% and boosted average revenue per user by 19.5% for high-risk segments.
- Developed a time series analysis module using Facebook's Prophet and ARIMA models. Forecasted user engagement metrics such as transaction frequency and app usage time with 92% accuracy over 30 days, aiding the churn prediction model.
- Implemented a customer segmentation system using K-means clustering and PCA in scikit-learn. Processed 50+ user attributes to identify 5 distinct personas, enhancing the effectiveness of retention campaigns by 35%.
- Built a churn prediction model with XGBoost, incorporating time series forecasts and customer segmentation features. Achieved 88% accuracy in predicting churn 60 days in advance, enabling proactive retention measures. Utilized MLflow for experiment tracking and model versioning, improving model iteration speed by 53%.
- Implemented an A/B testing framework using Python's Statsmodels, optimizing UI elements, notification strategies, and reward structures within the PayEase app.
- Created interactive Tableau dashboards for real-time visualizations of user segments, churn risk factors, and campaign effectiveness. Improved data-driven decision-making across departments by 43%, as measured by internal metrics.

EDUCATION

Masters Science in Information Technology - Indiana Wesleyan University, IN, USA

Bachelors of Computers - Satavahana University, India

CERTIFICATION

Microsoft Certified: Power BI Data Analyst Associate